FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation document. Use it as a model to help you write a complete, well-organised answer to the driving question. Read each section carefully, study the exemplar responses, and use them as a guide when you write your own answer. Your goal is to show that you truly understand how quadratic equations connect to real-life situations — not just how to solve them on paper. Write in full sentences, use correct mathematical vocabulary, and always check whether your answer makes sense in context.

Section 1: Identify the Real-Life Situation and Define the Unknown	
Describe the real-life problem in your own words. What quantity is unknown? How do you assign a variable to represent it? Why can't you solve this problem with simple division or a one-step calculation?	Achieng's school in Kisumu needs to seat 120 guests in a rectangular arrangement of chairs. The number of columns exceeds the number of rows by 7, but neither the number of rows nor columns is known. Because the two unknowns are related — one is 7 more than the other — and because their product must equal 120, you cannot find the answer by simply dividing 120 by a single number. You need to represent one unknown with a variable. Let the number of rows = x. Then the number of columns = $x + 7$. This step is critical: choosing a clear variable definition keeps your equation meaningful and prevents confusion later. Whenever a problem involves two related quantities whose product (or sum of products) gives a fixed total, a single variable and a quadratic equation are the right tools to use.

Section 2: Write and Justify the Quadratic Equation	
Write the equation that models the situation. Show every step of forming the equation. Explain why the equation contains a squared term — in other words, where does the 'quadratic' part come from in this real-life context?	Since rows × columns = total chairs, you can write: $x(x + 7) = 120$. Expanding the left side gives $x^2 + 7x = 120$, which rearranges to $x^2 + 7x - 120 = 0$. The squared term (x^2) appears because you are multiplying two expressions that both contain the variable x — rows and columns are each described using x, so their product naturally produces x^2. This is what makes the equation quadratic rather than linear. A linear equation would appear if only one dimension contained the unknown, but here both dimensions depend on x, so the equation must have degree 2. Recognising this structure is the key insight: any time two related lengths, quantities, or dimensions are multiplied together to give a fixed area or total, a quadratic equation is the correct model.

Section 3: Solve the Quadratic Equation and Interpret Both Roots	
Solve your equation using an appropriate method (factorisation, completing the square, or the quadratic formula). Show all working clearly. You will get two solutions — explain what each solution means in the real-life context and decide which solution is valid.	Starting from $x^2 + 7x - 120 = 0$, you look for two numbers that multiply to −120 and add to +7. Those numbers are +15 and −8, so the equation factorises as $(x + 15)(x - 8) = 0$. Setting each factor equal to zero gives $x = -15$ or $x = 8$. Now you must interpret both roots in context. $x = -15$ means there would be −15 rows of chairs. A negative number of rows is physically impossible — you cannot have a negative count of chairs in a hall in Kisumu or anywhere else. Therefore $x = -15$ is rejected. $x = 8$ is positive and realistic: 8 rows of chairs, and $8 + 7 = 15$ columns of chairs. Checking: $8 \times 15 = 120$ ✓. The valid solution is 8 rows and 15 columns. This step — checking both roots against the real-life constraints — is just as important as the algebra itself. In many practical problems, one root will be mathematically correct but contextually meaningless.

Section 4: Connect the Mathematics to the Broader Driving Question	
Go beyond Achieng's seating problem. Explain how the same process — defining a variable, writing a quadratic equation, solving it, and interpreting the roots — applies to at least ONE other real-life situation in Kenya. Use a specific, detailed example.	The same quadratic thinking applies to many everyday situations across Kenya. Consider a maize farmer in the Rift Valley who wants to enclose a rectangular shamba using 54 metres of fencing, and needs the enclosed area to be exactly 180 m². If the width of the shamba is w metres, then the length is $(27 - w)$ metres (since $2w + 2l = 54$, so $l = 27 - w$). The area condition gives $w(27 - w) = 180$, which expands to $27w - w^2 = 180$, or $w^2 - 27w + 180 = 0$. Factorising: $(w - 12)(w - 15) = 0$, giving $w = 12$ or $w = 15$. Both roots are positive and realistic here — a 12 m × 15 m shamba or a 15 m × 12 m shamba are the same rectangle described from different sides. Both roots are valid, which shows that context determines whether one or both solutions are accepted. In each case, the process is identical: name the unknown, form the product, expand, rearrange to standard form $ax^2 + bx + c = 0$, solve, and interpret. Quadratic equations are therefore a powerful and flexible tool for any Kenyan context involving areas, arrangements, or two related quantities multiplied together.

Section 5: Reflect on Your Learning and the Power of Quadratic Equations	
Look back at the full sequence of lessons. What was confusing at the start that now makes sense? What is the most important idea you are taking away about quadratic equations? How has your thinking changed from Lesson 1 to now?	At the start of this sequence, Achieng's problem seemed confusing — it was hard to see why a seating arrangement would produce an equation with x^2. Now it is clear: whenever two unknown but related quantities are multiplied together, the result is always a quadratic expression. The most important idea to take away is that quadratic equations are not just abstract algebra — they are a language for describing real situations involving area, dimensions, and arrangements. A second key insight is that getting two solutions is not a mistake; it is a feature of quadratic equations, and your job is always to test both solutions against the real-life context to decide which one (or sometimes both) make sense. Finally, the standard form $ax^2 + bx + c = 0$ is a tool: rearranging every quadratic into this form before solving makes it easier to choose the right method — factorisation when the numbers work out neatly, completing the square when they do not, or the quadratic formula when you need a guaranteed method. These skills transfer directly to future topics such as quadratic graphs, projectile motion in Physics, and financial calculations in Business Studies.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Defining the Unknown and Setting	Clearly and correctly defines the variable	Correctly defines the variable and links it	Attempts to define a variable but the

Up the Variable	with explicit units or context (e.g., 'Let x = the number of rows of chairs'). Explains in precise language why a single variable is sufficient to describe both unknowns and why simple division cannot solve the problem.	to the second unknown $(x + 7)$. Mentions why the problem cannot be solved by division alone, though the explanation may lack full depth.	definition is vague, incorrect, or disconnected from the context. May attempt to solve by trial and error without forming an equation.
Forming and Justifying the Quadratic Equation	Accurately writes $x(x + 7) = 120$, expands fully, and rearranges to standard form $x^2 + 7x - 120 = 0$ with all steps shown. Clearly explains that the squared term arises from multiplying two expressions both containing x, and uses the term 'quadratic' correctly.	Correctly forms and expands the equation to standard form with minor arithmetic slips that do not affect understanding. Gives a reasonable explanation of why the equation is quadratic.	Sets up the equation with errors (e.g., writes $x + (x + 7) = 120$ instead of a product) or fails to rearrange to standard form. Explanation of the squared term is absent or incorrect.
Solving the Equation and Interpreting Both Roots	Solves the equation correctly using a valid method (factorisation, completing the square, or quadratic formula) with all working shown. Explicitly interprets both roots in context, clearly explains why $x = -15$ is rejected, verifies $x = 8$ by substitution ($8 \times 15 = 120$), and states the final answer in a full sentence.	Obtains both correct roots and identifies the valid solution. Explains that a negative value is not meaningful in this context, though the verification step may be missing or incomplete.	Solves the equation with significant errors, or obtains only one root. Does not interpret the roots in context, or accepts $x = -15$ as a valid answer without question.
Applying the Quadratic Process to a New Kenyan Context	Presents a detailed, accurate, and fully worked second example rooted in a Kenyan context (e.g., a Rift Valley shamba, a Mombasa fishpond, a Nairobi market stall). Correctly forms, solves, and interprets the quadratic equation for the new situation, and comments meaningfully on whether one or both roots are valid.	Provides a relevant Kenyan example and forms a correct or near-correct quadratic equation. Solves the problem and gives a contextual interpretation, though the working or explanation may have minor gaps.	Provides an example that is vague, non-Kenyan, or incorrectly modelled. The quadratic equation formed is inaccurate, or the student cannot solve or interpret it in context.
Mathematical Communication and Reflection	Writes in clear, full sentences throughout. Uses correct mathematical vocabulary consistently (quadratic equation, standard form, roots, factorisation, variable, coefficient, valid/invalid solution). The reflection section shows genuine growth in thinking, identifies a specific misconception that was resolved, and connects quadratic equations to at least one other subject or future topic.	Uses mostly correct vocabulary and writes in reasonably clear sentences. The reflection identifies something that was initially confusing and explains the key learning, though connections to other subjects or future topics may be superficial.	Mathematical vocabulary is limited, missing, or used incorrectly. The reflection is brief, generic ('I learned how to solve quadratic equations'), or does not connect to the phenomenon or to broader applications.